

Dialysis Patient Management System

Data Collection · Insights · System Features

1. Data Collection

Collect all patient-related data in one centralized system to support patient monitoring, data analysis, reporting, and future AI-based insights.

1.1 Patient Profile

Recorded once when creating the patient's record and updated whenever necessary.

- Patient ID
- Full Name
- Age
- Gender
- Height
- Dry Weight
- Dialysis Start Date
- Diagnosis
- Comorbidities (Diabetes, Hypertension, etc.)
- Viral Status (HBV / HCV / HIV)
- Vascular Access (AV Fistula / Catheter / Graft)
- Allergies
- Current Medications
- Medication History

1.2 Dialysis Session Data

Before the Session

- Date
- Time
- Pre-Dialysis Weight
- Blood Pressure
- Heart Rate (Pulse)
- Body Temperature
- Current Symptoms (Dizziness, Shortness of Breath, Edema, etc.)

During the Session

- Blood Pressure Monitoring
- Heart Rate
- Ultrafiltration (UF) Volume
- UF Rate
- Blood Flow Rate

- Dialysate Flow Rate
- Medications Administered During Dialysis
- Session Complications (Hypotension, Muscle Cramps, Fainting, etc.)

After the Session

- Post-Dialysis Weight
- Blood Pressure
- Heart Rate
- Actual Session Duration
- Session Completion Status
- Nurse Notes

1.3 Periodic Laboratory Tests

- Hemoglobin
- Creatinine
- Urea
- Potassium
- Sodium
- Calcium
- Phosphorus
- Albumin
- Ferritin
- PTH
- CRP

For each laboratory test:

- Test Date
- Test Result
- Measurement Unit (mg/dL, mmol/L, etc.)

2. Insights

Transform raw patient data into meaningful information that supports doctors in clinical decision-making, helps administrators evaluate the dialysis unit, and provides simplified health summaries for patients.

2.1 Patient Insights

Patient Summary

- Is the patient's condition stable?
- Has the patient's condition improved?
- Is there any deterioration?
- Are there any important alerts?

Trend Analysis

Analyze patient data over time. Examples:

- Weight Trend
- Blood Pressure Trend
- Hemoglobin Trend
- Potassium Trend
- Albumin Trend
- Other Laboratory Test Trends

Time comparison options:

- Last Month
- Last 3 Months
- Last 6 Months
- Last Year

Session Analysis

- Total Dialysis Sessions
- Attendance Rate
- Completed Sessions Percentage
- Incomplete Sessions Percentage
- Number of Recorded Complications

Medication Tracking

- Medication History
- Latest Prescription Changes
- Treatment Progress Over Time

2.2 Dialysis Unit Insights

- Total Number of Patients
- Total Dialysis Sessions
- Average Session Duration
- Average Hemoglobin Level
- Average Potassium Level
- Number of Patients Outside the Normal Laboratory Range

- Number of Hypotension Cases
- Number of Muscle Cramp Cases
- Number of Fainting Cases
- Percentage of Sessions with Complications
- Machine Utilization Rate (if machine integration is available)

2.3 Patient Application Insights

Examples of patient-friendly messages:

- Your condition is stable.
- Your condition has improved compared to last month.
- Please continue following your medication and dialysis schedule.
- Some changes have been detected and will be reviewed by your doctor.
- Please discuss your latest results with your doctor during your next visit.

3. System Features

Define the core functionalities provided by the system for all users.

3.1 Patient Management

- Create a patient medical record.
- Update patient information.
- Manage patient medical history.
- Maintain patient records.

3.2 Dialysis Session Management

- Create a new dialysis session.
- Record pre-session data.
- Record intra-session data.
- Record post-session data.
- Record dialysis complications.
- Record ultrafiltration volume.
- Store complete dialysis session history.

3.3 Laboratory Management

- Add laboratory test results.
- Update laboratory results.
- View laboratory history.
- Compare laboratory results across different time periods.
- Link laboratory results to the patient's medical record.

3.4 Medication Management

- Add medications.
- Update prescriptions.
- Store medication history.
- Manage current treatment plans.

3.5 Monitoring & Tracking

- Monitor patient weight.
- Monitor blood pressure.
- Monitor laboratory results.
- Monitor dialysis complications.
- Monitor attendance and compliance.
- Track patient progress over time.

3.6 Analytics & Reporting

- Interactive Charts
- Trend Analysis
- Time-Based Comparisons

- Patient Reports
- Dashboards
- Key Performance Indicators (KPIs)

3.7 Notification & Alert System

- Dialysis session reminders.
- Medication reminders.
- Alerts for abnormal laboratory results.
- Complication alerts.
- Notifications for doctors and nurses when patient follow-up is required.

3.8 Dashboard Management

- Doctor Dashboard
- Nurse Dashboard
- Patient Mobile Application
- Admin / Unit Manager Dashboard

3.9 AI Features

- Risk Prediction
- Patient Risk Score
- Pattern Detection
- Smart Patient Summary
- Recommendation System
- Early Warning System